

CSE110 Fall 2025 Lab Exam - 04 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class LabExam4SetA{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] basket_capacity = {5, 6, 7, 2, 3};

        System.out.print("Enter the number of fruit types: ");
        int n = sc.nextInt();
        int[] fruits = new int[n];

        System.out.println("Enter the quantity of each fruit
type:");
        for (int i = 0; i < n; i++) {
            System.out.print("For Fruit type " + (i + 1) + " :
");
            fruits[i] = sc.nextInt();
        }

        System.out.println("Initial basket capacities:");
        for (int c = 0; c < basket_capacity.length; c++ ) {
            System.out.print(basket_capacity[c] + " ");
        }
        System.out.println();

        for (int i = 0; i < n; i++) {
            int fruit = fruits[i];
            boolean placed = false;

            for (int j = 0; j < basket_capacity.length; j++) {
                if (basket_capacity[j] >= fruit) {
                    basket_capacity[j] -= fruit;
                    placed = true;
                    break;
                }
            }

            if (!placed) {
                System.out.println("Fruit type " + (i + 1) + "
```

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```
with quantity " + fruit + " could not be placed.");
    }
}

System.out.println("Updated basket capacities:");
for (int c = 0; c < basket_capacity.length; c++) {
    System.out.print(basket_capacity[c] + " ");
}
}
```

RUBRIC

SL	Criteria	Marks
1	Correctly takes the number of fruit types as input, creates the array and reads the quantity of each fruit type into an array.	2
2	Prints the initial basket capacities correctly before allocation.	0.5
3	Accurately assigns each fruit type to the first basket with sufficient capacity; skips baskets that cannot accommodate the fruit.	3
4	Correctly reduces the basket capacity by the number of fruits allocated.	1.5
5	Identifies and displays fruit types that could not be placed, along with their quantities.	2
6	Prints the updated capacities of all baskets after allocation.	1

[SET-B]

SOLUTION

```
import java.util.Scanner;

public class LabExam4SetB{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of students present: ");
        int n = sc.nextInt();
        int[] roll = new int[n];
        System.out.println("Enter roll numbers:");
        for (int i = 0; i < n; i++) {
            System.out.print("Student " + (i + 1) + ": ");
            roll[i] = sc.nextInt();
        }
        System.out.println("\nOriginal attendance list:");
        for (int i = 0; i < n; i++) {
            System.out.print(roll[i] + " ");
        }
        System.out.println();
        int evenCount = 0, oddCount = 0;
        int highest = roll[0];
        int lowest = roll[0];
        for (int i = 0; i < n; i++) {
            if (roll[i] % 2 == 0)
                evenCount++;
            else
                oddCount++;
            if (roll[i] > highest)
                highest = roll[i];
            if (roll[i] < lowest)
                lowest = roll[i];
        }
        int[] reordered = new int[n];
        int index = 0;
        for (int i = 0; i < n; i++) {
            if (roll[i] % 2 == 0) {
                reordered[index++] = roll[i];
            }
        }
        for (int i = 0; i < n; i++) {
            if (roll[i] % 2 != 0) {
                reordered[index++] = roll[i];
            }
        }
    }
}
```

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```
    }
    System.out.println("\nReordered attendance list:");
    for (int i = 0; i < n; i++) {
        System.out.print(reordered[i] + " ");
    }
    System.out.println();
    System.out.println("\nNumber of even roll numbers: " +
evenCount);
    System.out.println("Number of odd roll numbers: " +
oddCount);
    System.out.println("\nHighest roll number: " + highest);
    System.out.println("Lowest roll number: " + lowest);
    }
}
```

RUBRIC

SL	Points To Meet	Marks [10]
1	The program correctly accepts the number of students and their roll numbers, storing them in an array.	1
2	Displays the original roll numbers in the order entered.	0.5
3	Correctly counts the number of even and odd roll numbers.	1.5
4	Finds and displays the highest and lowest roll numbers correctly.	2.5
5	Reorders the array so all even numbers appear first, then odd numbers, preserving original order.	4
6	Displays the reordered list correctly.	0.5